



SOUTH DAKOTA MINES[®]

An engineering, science and technology university

Leslie A. Rose Department of Mechanical Engineering

**Heat Transfer Analysis
and Thermal Protection System Sizing
of the Artemis II Orion Re-entry Capsule**

Owen Asbridge

May 7th, 2026

Contents

Nomenclature	2
1 Introduction	4
2 Problem Setup	5
2.1 Reentry Capsule and Shock Layer	5
2.2 TPS Wall Model, Governing Equation, and Boundary Conditions	6
2.3 Material Properties	7
2.4 Assumptions	7
3 Solution Approach	8
3.1 Trajectory Parsing and Calculations	8
3.2 Freestream Properties from the 1976 Standard Atmosphere	8
3.3 Stagnation-Point Heat Flux Correlations	9
3.3.1 Fay–Riddell Equation	9
3.3.2 Sutton–Graves Equation	10
3.4 Shock Layer and Wall Property Evaluation	10
3.4.1 Post-Shock Edge Properties	10
3.4.2 Wall Conditions	10
3.4.3 Temperature-Dependent Air Properties	11
3.4.4 Stagnation and Wall Enthalpies	12
3.4.5 Dissociation Enthalpy	12
3.4.6 Stagnation Velocity Gradient	12
3.5 Radiation Equilibrium Iteration	12
3.6 Finite Difference Solver and TPS Thickness Optimization	13
3.6.1 Governing Equation and Boundary Conditions	13
3.6.2 Finite Difference Derivation	14
3.6.3 Bisection Optimization and Factor of Safety	14
4 Results and Discussion	15
4.1 TPS Design Thickness	15
4.2 Temperature Profile Through the Shield	15
4.3 Assumptions Critique and Limitations	16
5 Ethical Considerations	17
6 Conclusion and Recommendations	18
Appendix	19
B. AI Usage	20

Nomenclature

a	= local speed of sound, m/s
c_p	= specific heat at constant pressure, J/kg·K
F	= Fourier number, $\alpha \Delta t / \Delta x^2$
h	= enthalpy, J/kg
h_D	= dissociation enthalpy, J/kg
k	= thermal conductivity, W/m·K
K_{SG}	= Sutton–Graves constant, 1.7415×10^{-4}
L	= TPS thickness, m
Le	= Lewis number
M	= Mach number
P	= pressure, Pa
Pr	= Prandtl number
$\dot{q}_w$	= stagnation-point heat flux, W/m ²
R_E	= mean Earth radius, km
R_g	= specific gas constant for air, J/kg·K
R_n	= vehicle nose radius, m
Sc	= Schmidt number
T	= temperature, K
t	= time, s
V	= flow velocity, m/s
x	= depth coordinate through TPS, m
α	= thermal diffusivity, m ² /s
γ	= ratio of specific heats
ε	= surface emissivity
μ	= dynamic viscosity, kg/m·s
ρ	= density, kg/m ³
σ	= Stefan–Boltzmann constant, W/m ² ·K ⁴

Subscripts

$0e$	= stagnation condition at boundary layer edge
∞	= freestream condition
e	= boundary layer edge (post-shock) condition
w	= wall condition

i = spatial node index
 n = time step index

1 Introduction

When a spacecraft returns from the Moon, it enters Earth’s atmosphere at speeds up to approximately 11.2 km/s, compressing the air ahead of the vehicle so violently that a shock layer forms and heats to temperatures exceeding 10,000 K [3]. The resulting convective heat flux at the stagnation point (the location of maximum heating on the heat shield nose) is sufficient to destroy any unprotected structure within seconds. A thermal protection system (TPS) is therefore a fundamental requirement for crew survival, and sizing it correctly is critical: too thin risks structural failure; too thick adds mass and increases launch cost.

This report presents a stagnation-point aerothermal heating analysis and TPS thickness optimization for the NASA Artemis II crew module reentry. The reentry trajectory is parsed from the NASA Artemis II ephemeris [5], freestream conditions are evaluated using the US Standard Atmosphere 1976 model [6], and stagnation-point heat flux is computed via the Fay–Riddell [3] and Sutton–Graves [8] correlations. A radiation equilibrium iteration yields the time-varying outer wall temperature, which drives a one-dimensional explicit finite difference solver. A bisection search over TPS thickness then identifies the minimum Avcoat heat shield thickness that keeps the inner aluminum structure below 422 K, with a factor of safety of 1.5 applied to produce the final design thickness.

2 Problem Setup

2.1 Reentry Capsule and Shock Layer

Figure 1 shows the Orion crew module during atmospheric reentry. At lunar-return velocities of approximately 11.2 km/s, a strong detached bow shock forms ahead of the blunt heat shield. Across this shock, the freestream air is compressed and heated to temperatures exceeding 10,000 K, initiating molecular dissociation of oxygen and nitrogen. The stagnation point (the location directly in front of the nose, where the flow velocity is reduced to zero) experiences the highest convective heat flux on the entire vehicle and is therefore the thermally critical design location.

The connection between trajectory data and heat flux is direct. The NASA Artemis II ephemeris [5] provides position and velocity vectors at each point along the reentry trajectory, from which altitude and freestream speed are extracted at every time step. The US Standard Atmosphere 1976 model [6] is then used to return the freestream air density ρ_∞ , temperature T_∞ , pressure P_∞ , and dynamic viscosity μ_∞ at each altitude. The resulting Mach number determines whether the flow is supersonic or subsonic and therefore which heat flux correlation applies. These freestream conditions are the direct inputs to the stagnation-point heat flux calculation, establishing the time-varying thermal load $q_w(t)$ that the TPS must withstand throughout reentry.

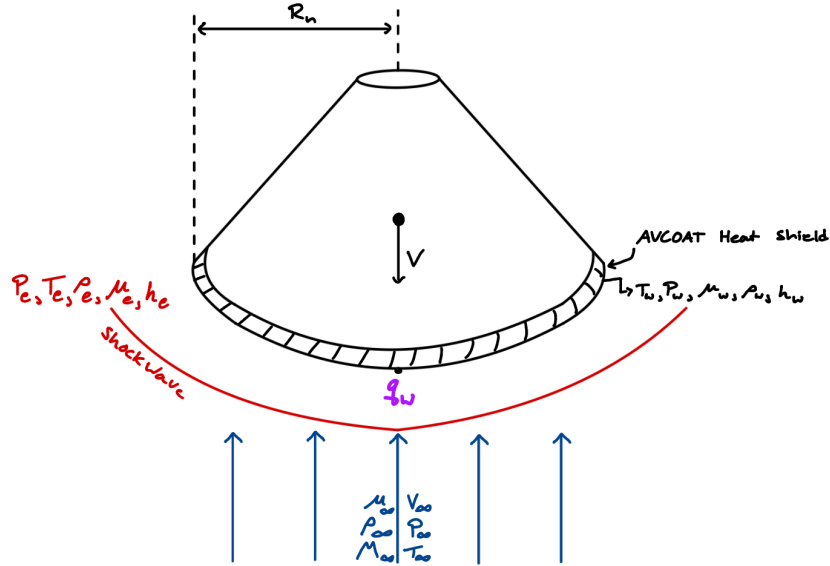


Figure 1: Orion crew module reentry schematic showing the bow shock, shock layer, and stagnation point at the heat shield nose.

2.2 TPS Wall Model, Governing Equation, and Boundary Conditions

Figure 2 shows the one-dimensional cross-section of the heat shield used in this analysis. The spatial coordinate x is measured inward from the outer surface, with $x = 0$ at the hot face exposed to the shock layer and $x = L$ at the inner face bonded to the aluminum crew module structure.

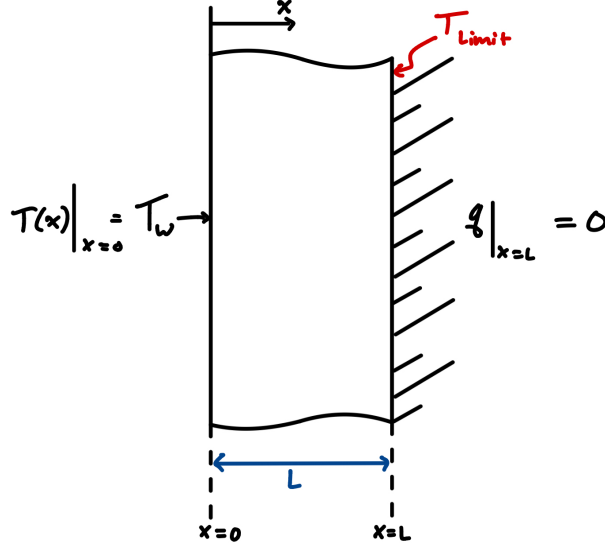


Figure 2: One-dimensional TPS wall cross-section showing the Avcoat slab, coordinate definition, boundary conditions, and inner aluminum structure.

Heat transfer through the TPS is governed by the one-dimensional transient heat conduction equation [2]:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad (1)$$

Initial condition: The shield is assumed to be at a uniform cold-wall temperature at the start of reentry:

$$T(x, 0) = 300 \text{ K} \quad \forall x \in [0, L] \quad (2)$$

Outer surface ($x = 0$) — prescribed wall temperature: The outer face simultaneously receives convective heat flux from the shock layer and re-radiates energy to space. The radiation equilibrium temperature $T_w(t)$ at which these are exactly balanced is derived in Section 4 and prescribed directly as a Dirichlet boundary condition:

$$T(0, t) = T_w(t) \quad (3)$$

Inner surface ($x = L$) — adiabatic: No heat is permitted to escape through the back face of the TPS, representing the most conservative possible assumption for inner wall

temperature rise:

$$\left. \frac{\partial T}{\partial x} \right|_{x=L} = 0 \quad (4)$$

2.3 Material Properties

Charred-state material properties for Avcoat 5026-39H/CG are used throughout the analysis [4]. Char has a lower thermal conductivity than virgin Avcoat, meaning less heat is conducted inward per unit time. Using charred-state properties therefore provides a conservative upper bound on the conducted heat load and the resulting inner wall temperature. All properties are listed in Table 1.

Table 1: Avcoat 5026-39H/CG Charred-State Material Properties [4]

Property	Symbol	Value	Units
Thermal conductivity	k	0.242	W/m·K
Density	ρ	264	kg/m ³
Specific heat	c_p	2,740	J/kg·K
Surface emissivity	ε	0.49	—

The Orion capsule nose radius is taken as $R_n = 2.5$ m. This parameter enters the stagnation velocity gradient term in the heat flux correlation — a larger nose radius reduces the gradient, and therefore lowers the peak heat flux. The role of R_n in the governing equations is discussed in detail in Section 4.

2.4 Assumptions

1. **One-dimensional conduction:** Valid at the stagnation point where lateral surface temperature gradients are negligible compared to depth-wise gradients through the shield.
2. **Constant charred-state material properties:** Conservative; real Avcoat ablates and the char front recedes during reentry, but ablation is not modeled here.
3. **Adiabatic inner wall:** No heat escapes through the back face, the worst-case scenario for inner wall temperature rise.
4. **Calorically perfect gas with $\gamma = 1.4$:** Used for the Rankine–Hugoniot post-shock relations. Dissociation effects are captured separately through the Lewis number correction in the heat flux correlation rather than through a variable γ [7].
5. **Fully catalytic wall:** All dissociated species recombine at the surface, depositing their dissociation energy. This maximizes the computed heat flux and provides a conservative upper-bound estimate [3].

3 Solution Approach

3.1 Trajectory Parsing and Calculations

The NASA Artemis II ephemeris file provides position vectors (x, y, z) in feet and velocity vectors (v_x, v_y, v_z) in ft/s with respect to time of the crew capsule, expressed in the M50 inertial reference frame (Earth-Centered Inertial Frame). At each time step, altitude is derived by computing the radial distance from Earth's centre and subtracting the mean Earth radius $R_E = 6,371$ km:

$$h = \sqrt{x^2 + y^2 + z^2} - R_E \quad (5)$$

The scalar freestream speed is the Euclidean magnitude of the velocity vector:

$$V_\infty = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad (6)$$

Both quantities are converted to km and km/s respectively for subsequent calculations. The resulting trajectory is shown in Figure 3, which illustrates altitude decreasing from approximately 120 km and speed dropping from approximately 11 km/s over the roughly 800 s reentry window.

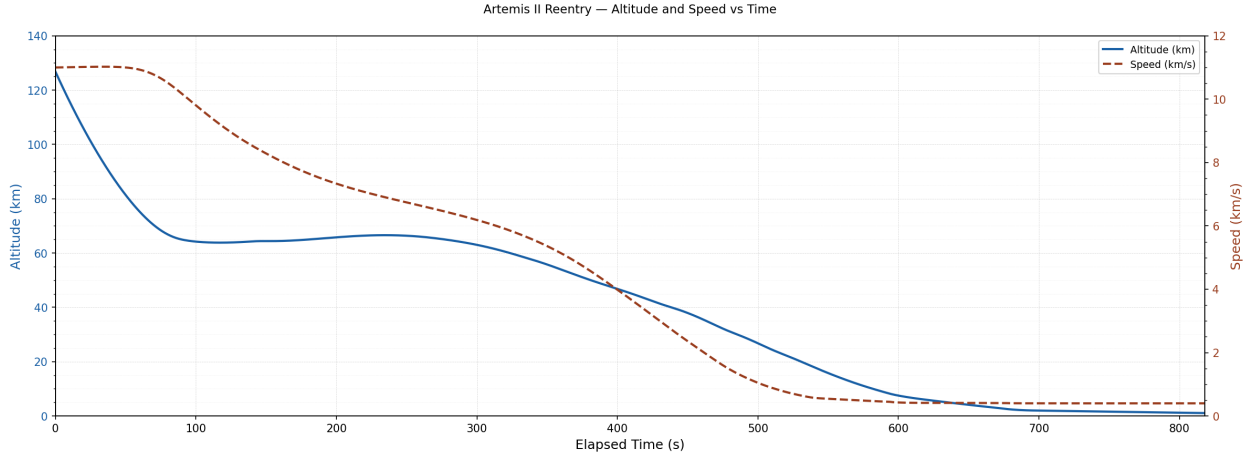


Figure 3: Artemis II reentry trajectory: altitude (left axis) and freestream speed (right axis) versus elapsed time, derived from the NASA Artemis II ephemeris.

3.2 Freestream Properties from the 1976 Standard Atmosphere

At each trajectory time step, the computed altitude is passed into the US Standard Atmosphere 1976 model [6] to return the freestream air density ρ_∞ , temperature T_∞ , pressure P_∞ , dynamic viscosity μ_∞ , and local speed of sound a . The freestream Mach number is the

$$M_\infty = \frac{V_\infty}{a} \quad (7)$$

The resulting freestream property histories are shown in Figure 4. Density and pressure span approximately seven orders of magnitude across the reentry corridor and are presented on a logarithmic scale. The non-monotonic temperature profile reflects the stratospheric temperature inversion characteristic of the Standard Atmosphere model. The Mach number derived from these properties is the key switching parameter in the heat flux calculation: for $M_\infty \geq 1$ the Fay–Riddell correlation is applied, and for $M_\infty < 1$ the Sutton–Graves correlation is used, as described in Section 3.3.

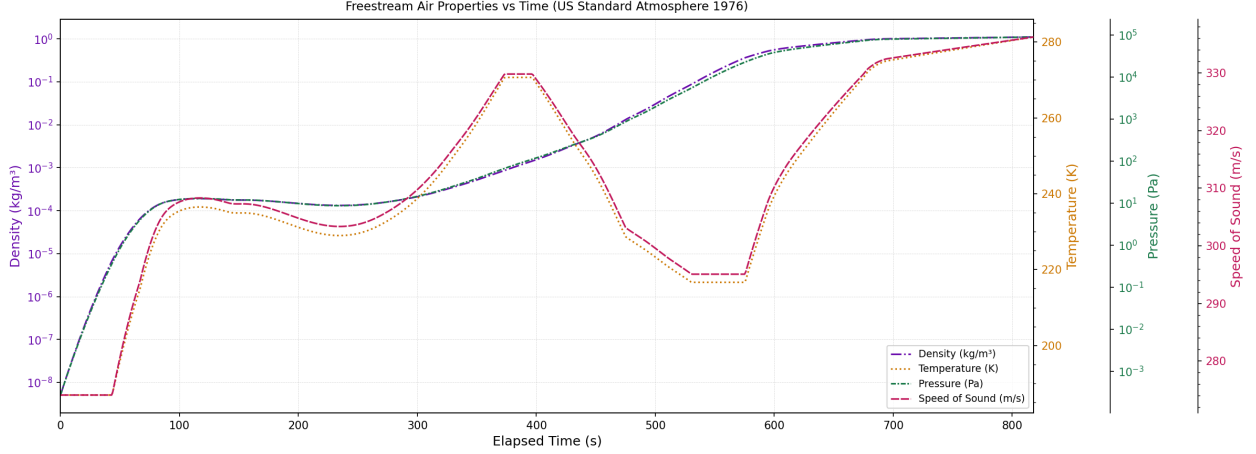


Figure 4: Freestream air properties along the reentry trajectory computed from the US Standard Atmosphere 1976 model: density, temperature, pressure, and speed of sound versus elapsed time.

3.3 Stagnation-Point Heat Flux Correlations

Two correlations are used to compute the stagnation-point convective heat flux, selected based on the local Mach number at each trajectory point. For $M_\infty \geq 1$ the Fay–Riddell correlation is applied, as the Rankine–Hugoniot shock relations are physically valid in this regime. For $M_\infty < 1$, a bow shock no longer exists and the Sutton–Graves empirical correlation is used as a fallback; heat flux in this subsonic phase is several orders of magnitude below the peak and contributes negligibly to the total heat load.

3.3.1 Fay–Riddell Equation

The stagnation-point heat flux for $M_\infty \geq 1$ is given by the Fay–Riddell equation [3]:

$$\dot{q}_w = 0.763 Pr^{-0.6} (\rho_e \mu_e)^{0.4} (\rho_w \mu_w)^{0.1} \sqrt{\left(\frac{dV}{dx}\right)_s} (h_{0e} - h_w) \left[1 + (Le^{0.52} - 1) \frac{h_D}{h_{0e}}\right] \quad (8)$$

The physical role of each factor is as follows:

- $Pr^{-0.6}$: correction for the ratio of momentum to thermal diffusivity in the boundary layer.

- $(\rho_e \mu_e)^{0.4}$: dominant transport property weighting; edge conditions drive the heat transfer.
- $(\rho_w \mu_w)^{0.1}$: weak wall transport property correction.
- $\sqrt{(dV/dx)_s}$: stagnation velocity gradient, encoding the nose radius and post-shock pressure effect.
- $(h_{0e} - h_w)$: enthalpy driving force between the hot shock layer and the cooler wall.
- $[1 + (Le^{0.52} - 1) h_D/h_{0e}]$: dissociation correction. For a fully catalytic wall with $Le > 1$, recombination energy adds meaningfully to the total heat flux.

This correlation requires properties at the boundary layer edge (immediately behind the normal shock) and at the wall, which are derived in full in Section 3.4.

3.3.2 Sutton–Graves Equation

The Sutton–Graves empirical correlation [8] relates heat flux directly to freestream density, nose radius, and the cube of velocity:

$$\dot{q}_w = K_{SG} \sqrt{\frac{\rho_\infty}{R_n}} V_\infty^3 \quad K_{SG} = 1.7415 \times 10^{-4} \quad (9)$$

3.4 Shock Layer and Wall Property Evaluation

3.4.1 Post-Shock Edge Properties

The boundary layer edge properties are taken as the post-normal-shock state, computed via the Rankine–Hugoniot relations for a calorically perfect gas with $\gamma = 1.4$ [7]:

$$\frac{T_e}{T_\infty} = \left[1 + \frac{2\gamma(M_\infty^2 - 1)}{\gamma + 1} \right] \left[\frac{2 + (\gamma - 1)M_\infty^2}{(\gamma + 1)M_\infty^2} \right] \quad (10)$$

$$\frac{P_e}{P_\infty} = \left[1 + \frac{2\gamma}{\gamma + 1} (M_\infty^2 - 1) \right] \quad (11)$$

$$\frac{\rho_e}{\rho_\infty} = \frac{(\gamma + 1)M_\infty^2}{2 + (\gamma - 1)M_\infty^2} \quad (12)$$

3.4.2 Wall Conditions

There is no pressure gradient normal to the wall, so the wall pressure equals the edge pressure. The wall density follows from the ideal gas law:

$$P_w = P_e \quad (13)$$

$$\rho_w = \frac{P_w}{R_g T_w} \quad (14)$$

where $R_g = 287.1$ J/kg·K is the specific gas constant for air.

3.4.3 Temperature-Dependent Air Properties

The temperatures encountered behind the bow shock require temperature-dependent property models, as constant-property assumptions are invalid in this regime.

Dynamic Viscosity. Dynamic viscosity at both the boundary layer edge and the wall is evaluated using Sutherland's law, with the freestream pair (μ_∞, T_∞) as the reference [2]:

$$\mu(T) = \mu_\infty \left(\frac{T}{T_\infty} \right)^{3/2} \frac{T_\infty + S}{T + S} \quad S = 110.4 \text{ K} \quad (15)$$

Specific Heat. The specific heat c_p varies significantly with temperature due to vibrational mode excitation above 800 K and the onset of molecular dissociation above 2,000 K [3]. The following piecewise model is used:

$$c_p(T) = \begin{cases} 1005 & T < 800 \text{ K} \\ 1005 + 0.200 (T - 800) & 800 \leq T < 2000 \text{ K} \\ 1245 + 0.150 (T - 2000) & 2000 \leq T < 4000 \text{ K} \\ 1545 + 0.050 (T - 4000) & T \geq 4000 \text{ K} \end{cases} \quad [\text{J/kg}\cdot\text{K}] \quad (16)$$

Thermal Conductivity. Thermal conductivity of air is evaluated using a Sutherland-form relation with a separate constant appropriate for conductivity [2]:

$$k_{air}(T) = k_{ref} \left(\frac{T}{T_{ref}} \right)^{3/2} \frac{T_{ref} + S_k}{T + S_k} \quad (17)$$

where $k_{ref} = 0.02624 \text{ W/m}\cdot\text{K}$ at $T_{ref} = 300 \text{ K}$ and $S_k = 194 \text{ K}$.

Prandtl Number. With viscosity, specific heat, and thermal conductivity defined, the Prandtl number at the boundary layer edge is:

$$Pr = \frac{\mu_e c_{p,e}}{k_{air,e}} \quad (18)$$

Lewis Number. The Lewis number governs the relative contribution of mass diffusion and thermal diffusion to the stagnation-point heat flux, expressed in terms of the Schmidt number $Sc = 0.75$, treated as constant for air:

$$Le = \frac{Sc}{Pr} \quad (19)$$

At typical reentry conditions $Le \approx 1.4$, meaning mass diffusion slightly exceeds thermal diffusion and the dissociation correction term in the Fay–Riddell equation makes a meaningful contribution to the total heat flux.

3.4.4 Stagnation and Wall Enthalpies

The total stagnation enthalpy at the boundary layer edge includes both thermal and kinetic energy. At the stagnation point essentially all kinetic energy is converted to thermal energy [3]:

$$h_{0e} = c_{p,e} T_e + \frac{1}{2} V_\infty^2 \quad (20)$$

The wall enthalpy is evaluated under calorically perfect gas assumptions:

$$h_w = c_{p,w} T_w \quad (21)$$

3.4.5 Dissociation Enthalpy

For a fully catalytic wall, all dissociated species recombine at the surface and deposit their dissociation energy. The mass-weighted dissociation enthalpy of air, composed of 21% O₂ and 79% N₂ by mole fraction, is [3]:

$$h_D = 0.21 h_{D,O_2} + 0.79 h_{D,N_2} = 0.21 \times 15.576 \times 10^6 + 0.79 \times 33.550 \times 10^6 = 29.78 \text{ MJ/kg} \quad (22)$$

3.4.6 Stagnation Velocity Gradient

The stagnation velocity gradient for Newtonian hypersonic flow is [3]:

$$\left(\frac{dV}{dx} \right)_s = \frac{1}{R_n} \sqrt{\frac{2(P_e - P_\infty)}{\rho_e}} \quad (23)$$

This term shows explicitly how nose radius controls heating: a larger R_n reduces the velocity gradient and therefore lowers the peak heat flux, scaling as $R_n^{-1/2}$.

3.5 Radiation Equilibrium Iteration

The outer TPS surface reaches a radiation equilibrium temperature at which the incoming convective flux is exactly balanced by re-radiation to space. Applying the Stefan–Boltzmann law gives [2]:

$$\dot{q}_w = \varepsilon \sigma T_w^4 \quad \Rightarrow \quad T_w = \left(\frac{\dot{q}_w}{\varepsilon \sigma} \right)^{1/4} \quad (24)$$

This is not a simple explicit solve — the Fay–Riddell equation requires T_w to evaluate h_w and ρ_w , making the system circular. Fixed-point iteration resolves this at every trajectory time step:

1. Initialise $T_w^{(0)} = 300 \text{ K}$ (cold wall) at all trajectory points.
2. Compute $\dot{q}_w^{(k)}$ via Eq. (8) using the current $T_w^{(k)}$.

3. Update the wall temperature from radiation equilibrium:

$$T_w^{(k+1)} = \left(\frac{\dot{q}_w^{(k)}}{\varepsilon \sigma} \right)^{1/4} \quad (25)$$

4. Repeat until $\max |T_w^{(k+1)} - T_w^{(k)}| < 0.1$ K across all trajectory points.

Convergence is typically achieved within 10–15 iterations. The converged heat flux and wall temperature histories are shown in Figure 5. Heat flux peaks sharply mid-trajectory, near 60–70 km altitude, where the product $\rho_\infty V_\infty^3$ is maximised.

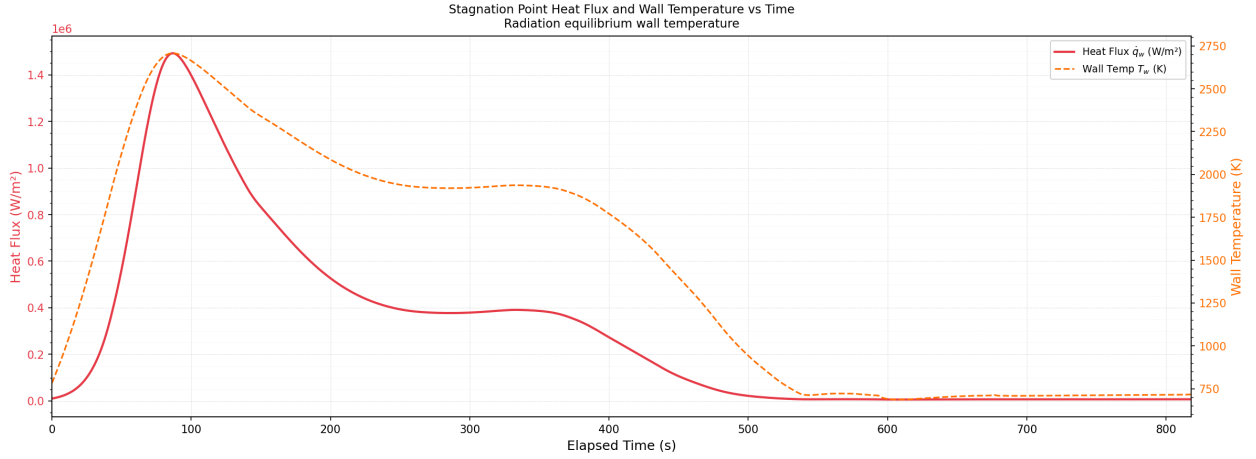


Figure 5: Converged stagnation-point heat flux $\dot{q}_w$ (left axis) and radiation equilibrium outer wall temperature T_w (right axis) versus elapsed time.

3.6 Finite Difference Solver and TPS Thickness Optimization

3.6.1 Governing Equation and Boundary Conditions

The governing equation, initial condition, and boundary conditions established in Section 2 are restated here for clarity before their numerical implementation:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad x \in [0, L], \quad t > 0 \quad (26)$$

$$T(x, 0) = 300 \text{ K} \quad \forall x \in [0, L] \quad (27)$$

$$T(0, t) = T_w(t) \quad (28)$$

$$\left. \frac{\partial T}{\partial x} \right|_{x=L} = 0 \quad (29)$$

The objective is to find the minimum thickness L^* such that $T(L, t) \leq T_{limit} = 422$ K for all t throughout the reentry event, with a factor of safety $FS = 1.5$ applied to the result.

3.6.2 Finite Difference Derivation

The continuous derivatives in Eq. (26) are approximated using finite difference expressions on a uniform spatial grid of $N = 50$ nodes with spacing $\Delta x = L/(N - 1)$ and discrete time steps Δt . The node index is i and the time step index is n .

Time derivative — forward difference:

$$\frac{\partial T}{\partial t} \approx \frac{T_i^{(n+1)} - T_i^{(n)}}{\Delta t} \quad (30)$$

Spatial second derivative — central difference:

$$\frac{\partial^2 T}{\partial x^2} \approx \frac{T_{i+1}^{(n)} - 2T_i^{(n)} + T_{i-1}^{(n)}}{\Delta x^2} \quad (31)$$

Substituting Eqs. (30) and (31) into Eq. (26) and defining the Fourier number $F = \alpha \Delta t / \Delta x^2$, the interior node update equation for $i = 1, \dots, N - 2$ is:

$$T_i^{(n+1)} = T_i^{(n)} + F \left(T_{i+1}^{(n)} - 2T_i^{(n)} + T_{i-1}^{(n)} \right) \quad (32)$$

Outer surface node ($i = 0$). Equation (28) prescribes the temperature directly, so no update equation is needed:

$$T_0^{(n+1)} = T_w(t^{(n)}) \quad (33)$$

Inner wall node ($i = N - 1$). The adiabatic condition is discretised by introducing a ghost node at $i = N$ with $T_N^{(n)} = T_{N-2}^{(n)}$, giving:

$$T_{N-1}^{(n+1)} = T_{N-1}^{(n)} + 2F \left(T_{N-2}^{(n)} - T_{N-1}^{(n)} \right) \quad (34)$$

3.6.3 Bisection Optimization and Factor of Safety

The minimum TPS thickness L^* is found by bisection search between $L_{min} = 5$ mm and $L_{max} = 300$ mm. At each candidate thickness the full finite difference solver is run over the entire reentry trajectory. The test condition is:

$$f(L) = \begin{cases} \text{too thin} & \text{if } (\max T_{N-1}(t)) > T_{limit} \\ \text{acceptable} & \text{otherwise} \end{cases} \quad (35)$$

where $T_{limit} = 422$ K, consistent with the 300°F inner structural wall limit [1]. Bisection continues until the bracket width is less than 1 mm, yielding L^* . A factor of safety $FS = 1.5$ is then applied:

$$L_{design} = FS \times L^* \quad (36)$$

4 Results and Discussion

4.1 TPS Design Thickness

The bisection optimization converged on a minimum compliant thickness of $L^* = 4.93$ cm, representing the thinnest Avcoat slab that keeps the inner aluminum wall below $T_{limit} = 422$ K throughout the entire reentry event. Applying the factor of safety $FS = 1.5$ per Eq. (36) yields the design thickness:

$$L_{design} = 1.5 \times 4.93 \text{ cm} = 7.40 \text{ cm} \quad (37)$$

4.2 Temperature Profile Through the Shield

Figure 6 shows the temperature distribution through the TPS heat shield as a function of depth and elapsed time at the design thickness $L_{design} = 7.40$ cm.

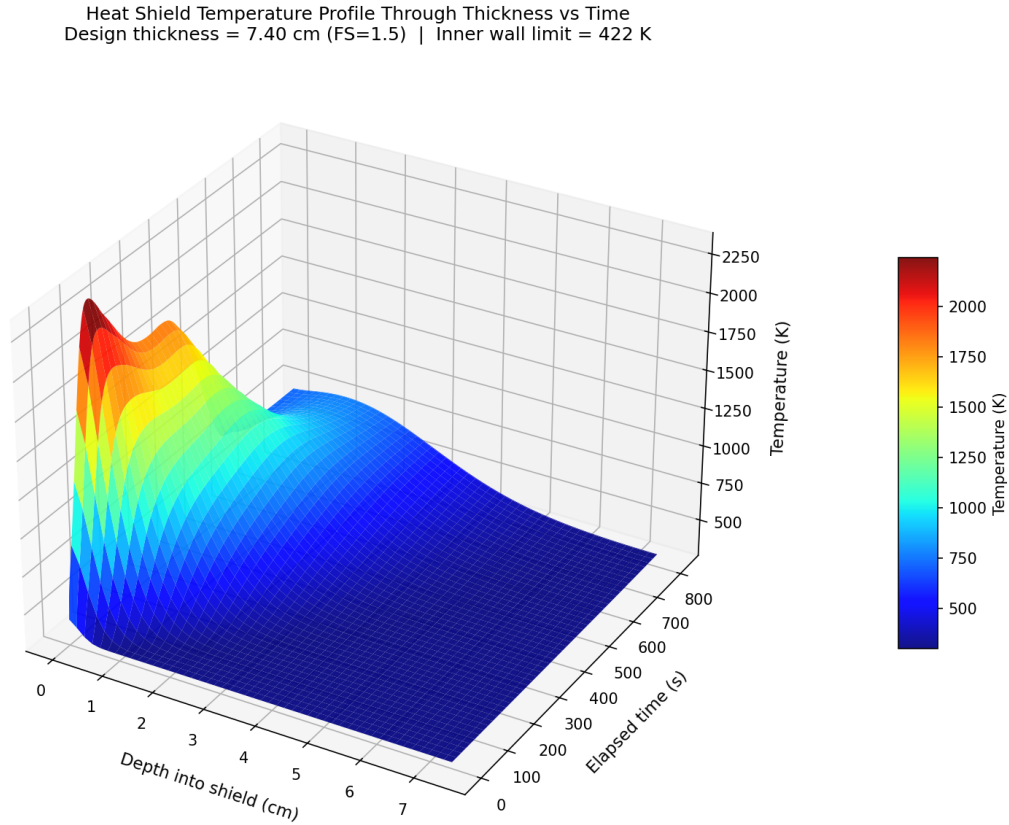


Figure 6: Temperature distribution through the Avcoat TPS heat shield versus depth and elapsed time.

The peak outer surface temperature reached 2,707.3 K, occurring at the time of maximum heat flux near 60–70 km altitude, consistent with the radiation equilibrium wall temperatures derived in Section 4. The outer surface temperature then decays rapidly as the vehicle decelerates into the lower atmosphere and heat flux subsides.

The peak inner wall temperature reached only 305.4 K — just 5.4 K above the initial cold-wall temperature of 300 K and well below the structural limit of $T_{limit} = 422$ K. The heat penetrates progressively deeper into the shield during reentry but reduces significantly before reaching the inner face. At the design thickness, the Avcoat heat shield acts as an effective thermal barrier, with the bulk of the temperature drop concentrated in the outer portion of the shield where the gradient is steepest. The large margin between the peak inner wall temperature (305.4 K) and the limit (422 K) reflects the applied factor of safety.

4.3 Assumptions Critique and Limitations

The most significant simplification in this analysis is the absence of ablation. Real Avcoat ablates during reentry, removing mass and carrying heat away from the surface through pyrolysis and char formation — an effect not captured here. As a result, this analysis over-predicts the heat load conducted into the shield, meaning the computed thickness is conservative.

The use of constant charred-state material properties throughout the reentry is a further simplification. In reality, the TPS transitions from virgin Avcoat to char as ablation proceeds, and the thermophysical properties change accordingly. Using charred-state properties from the outset further overestimates the insulating performance and reinforces the conservatism.

Finally, the Artemis II reentry trajectory includes a skip maneuver, in which the capsule enters the upper atmosphere and uses aerodynamic lift to briefly return to space before a final entry, as visible in Figure 3. During the pull-up and skip phases, the vehicle operates at a nonzero angle of attack, which redistributes the heating load across the surface. Under these conditions, the assumption that the geometric stagnation point governs the maximum heat flux may not hold. Resolving the full surface heat flux distribution over a skip trajectory would require computational fluid dynamics (CFD) applied to the complete vehicle geometry at each trajectory point.

5 Ethical Considerations

The analysis presented in this report directly governs whether astronauts survive reentry. An undersized thermal protection system is not a recoverable failure. At re-entry velocities, any breach of the crew module during peak heating is unsurvivable. This places a profound ethical responsibility on every engineer involved in TPS design.

History has demonstrated the cost of thermal protection failures. The Space Shuttle *Columbia* disaster in 2003 killed all seven crew members after foam debris damaged the heat shield, allowing superheated plasma to penetrate the wing structure during reentry [9]. The accident was attributed not only to a physical failure, but to organisational ones — schedule pressure, normalisation of known risks, and engineers unable to escalate safety concerns effectively.

The conservative assumptions adopted throughout this analysis (charred-state material properties, a fully catalytic wall, an adiabatic inner boundary, and a factor of safety of 1.5) exist precisely because the cost of being wrong is measured in human lives. Reducing these margins without rigorous justification and independent review would be unethical. Engineers working on crewed systems have a professional obligation to hold safety to the highest standard, to be transparent about the limitations of their analyses, and to resist pressure to approve a design they believe to be unsafe.

6 Conclusion and Recommendations

This report presented a stagnation-point aerothermal heating analysis and TPS thickness optimization for the Artemis II crew module reentry. The Fay–Riddell and Sutton–Graves correlations were applied along the reentry trajectory to produce a time-varying heat flux profile, peaking near 60–70 km altitude. A radiation equilibrium iteration yielded the outer wall temperature history, which drove a one-dimensional explicit finite difference solver. Bisection search identified a minimum compliant Avcoat thickness of $L^* = 4.93$ cm, with a factor of safety of 1.5 applied to give a final design thickness of $L_{design} = 7.40$ cm. Despite the outer surface reaching a peak temperature of 2,707.3 K, the inner wall peaked at only 305.4 K, just 5.4 K above the initial cold-wall temperature and well below the structural limit of 422 K. This contrast demonstrates the effectiveness of ablative TPS design: by absorbing and re-radiating the vast majority of the incoming flux at the outer surface, only a small fraction conducts inward, leaving the crew module structure at near-ambient temperature throughout reentry.

The following recommendations are made for future work:

- **Incorporate ablation modelling.** The absence of ablation is the single largest simplification in this analysis. Coupling a surface recession model to the heat conduction solver would produce a more realistic, and likely thinner, design thickness.
- **Apply variable material properties.** Replacing the constant charred-state properties with temperature-dependent virgin-to-char transition data would improve fidelity across the full reentry heating pulse.
- **Extend to full surface heating.** The skip reentry trajectory and nonzero angle of attack redistribute heating across the shield surface. A CFD analysis of the complete vehicle geometry would resolve whether the stagnation point truly governs the thermal design under these conditions.

Appendix

A. References

References

- [1] Ko, W. L., Gong, L., and Quinn, R. D., *Reentry Thermal Analysis of a Generic Crew Exploration Vehicle Structure*, NASA/TM-2007-214607, NASA Dryden Flight Research Center, Edwards, CA, May 2007.
- [2] Bergman, T. L., Lavine, A. S., Incropera, F. P., and DeWitt, D. P., *Fundamentals of Heat and Mass Transfer*, 7th ed., John Wiley & Sons, Hoboken, NJ, 2011.
- [3] Anderson, J. D., Jr., *Hypersonic and High-Temperature Gas Dynamics*, 2nd ed., AIAA Education Series, Reston, VA, 2006.
- [4] National Aeronautics and Space Administration, “Materials Database (TPSX),” NASA Ames Research Center, <https://tpsx.arc.nasa.gov/>, Accessed May 2026.
- [5] National Aeronautics and Space Administration, “Track NASA’s Artemis II Mission in Real Time,” NASA Web Portal, <https://www.nasa.gov/specials/track-artemis/>, 2024.
- [6] National Oceanic and Atmospheric Administration, *U.S. Standard Atmosphere, 1976*, NOAA-S/T-76-1562, Washington, D.C., 1976.
- [7] Anderson, J. D., Jr., *Fundamentals of Aerodynamics*, 5th ed., McGraw-Hill, New York, 2011.
- [8] Dec, J. A., “Aerothermodynamics Course,” *Thermal and Fluids Analysis Workshop (TFAWS 2012)*, NASA Marshall Space Flight Center, Huntsville, AL, August 2012. URL: <https://tfaws.nasa.gov/TFAWS12/Proceedings/Aerothermodynamics%20Course.pdf>.
- [9] Columbia Accident Investigation Board, *Report of the Columbia Accident Investigation Board*, Volume I, NASA, Washington, D.C., August 2003.

B. AI Usage

AI assistance was used in the production of this report in the following capacities.

Report Writing and Formatting. AI was used to format the $\text{\LaTeX}$ code, including equations, figures, and general document layout. It was also used to assist with rewording sentences for clarity, checking grammar and spelling, and verifying the accuracy of technical claims. Sources suggested by AI were independently reviewed for relevance and accuracy before inclusion.

Numerical Code Development. AI was used in the development of the finite difference solver. Specifically, it was consulted during the algorithm design phase, and once the approach was finalised, it was used to produce the initial implementation. All code was reviewed and validated against expected physical behaviour to confirm accuracy. AI was additionally used to format Python figures and code comments for clean, publication-quality output and to assist in debugging errors during development.